

Building Your Business

by Greg Thomas



Understanding how the local weather will create performance opportunities, combined with a cost-effective plan for acquiring customers, will steer you down the road to success.

Sadly, just going to the Affordable Comfort conference and learning the latest in building science, while fine-tuning one's technical skills, won't guarantee a successful building performance contracting business. In the 20 years that I have been working in this field, I have seen very technically talented building science believers lose a lot of money when they found out that consumers weren't knocking on their doors and their phones weren't ringing off the hook.

Yet there are building performance contractors who are running successful businesses. Just what are the successful business models for such contractors? Understanding the local market for comfort and energy improvements and knowing cost-effective ways to acquire customers are two essentials for creating a successful building performance business.

It's the Weather, Stupid

Success requires positioning yourself properly in your marketplace. To do that, you first have to understand how your local weather determines which building performance problems are most common in your area. Joe Istiburek, principal investigator for Building Science Corporation, has

identified different climate zones that affect building performance: Cold, Mixed, Hot-Humid, and Hot-Dry/Mixed-Dry (see "Opportunities by Climate Zone," p. 16). Each of these zones subject buildings to different combinations of temperature and humidity, and creates different kinds of performance problems.

A quick review of the opportunity list shows that HVAC problems tend to dominate in hot climates, and that envelope issues tend to dominate in cold climates. The more extreme the temperature or the humidity, the more severe the potential for performance problems. In truly extreme climates, many general contractors tend to know something about building science, because building failure is more problematic in these climates and the consequences can be so expensive. From Alaska to Saskatchewan to Florida, liability driven by extreme weather has given building science businesses a head start.

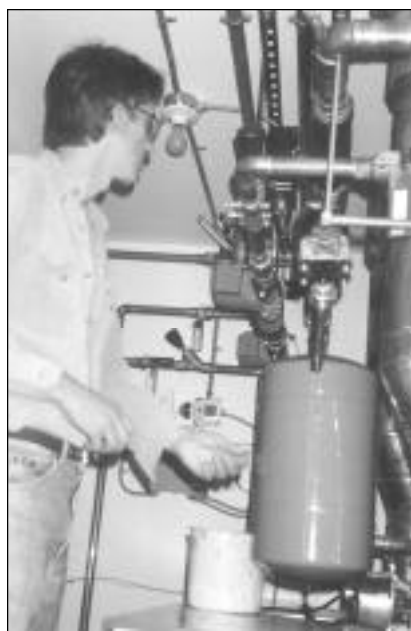
Geography and the weather have also influenced construction styles, and therefore the opportunities to solve problems. In hot climates, attic ductwork has become the standard, creating its own set of problems caused by duct leakage. In the north, ductwork is usually in the basement. Hot-climate residential construction often doesn't even have basements. Sometimes really big problems get their start when an unsuspecting architect or developer imports a construction style, which may have been creating minor problems in one climate zone, into a more severe zone.

Some markets are less dependent on the weather. Multifamily markets tend to be driven by the cost of energy and increasingly by the liability associated with indoor air quality, such as second-hand smoke migrating through buildings. A market with a large number of FHA-financed homes may create the critical mass necessary for setting up a home energy rater and energy-efficient

mortgage facilitator partnership. A rapidly growing market for new construction may drive home energy warranties and Energy Star labeling. And health and safety considerations such as backdrafting are universal.

Who Ya Gonna Call?

After you have determined the best services to offer in your area, you have to figure out how to get potential customers to identify you as the solution to



Greg Thomas inspects the boiler system at EcoVillage at Ithaca, a sustainable co-housing community. Customers, like the residents at EcoVillage, will often choose high performance, if given the choice.

their problems, and how to do this in a way that is cost effective. Costs involved in acquiring customers include advertising and marketing expenses, the cost of time spent on the phone convincing customers to buy your services, the cost of calculating an estimate, the cost of writing up a proposal, and the cost of any other tasks you have to do to get a signed contract. If you get work from 50% of the potential customers you interact with, that work has to carry the cost of preparing proposals for the other 50% who chose not to work with you. In business terms, the cost of acquiring the customer has to be offset by the gross margin on each job you sell. The smaller the job, the more jobs

you need to sell, and the more you need to have a very low cost means of acquiring customers.

The high cost of educating customers so that they will spend the extra money on the contractor with the diagnostic skills has been one of the biggest barriers to the growth of this industry. But this cost is unavoidable, no matter what type of building performance contractor you are or plan to be (see "Fitting into the Marketplace, p. 18). Building performance contractors can't compete on price against contractors who don't buy diagnostic equipment or attend training events.

The cost of acquiring customers is a particular problem for energy raters. If a rater must sell each customer, one by one, on the importance of getting a rating, the cost of acquiring customers is too high relative to the low gross margin on each job. To survive, most successful energy raters employ one of three strategies aimed at getting someone else to bear the costs of acquiring customers: They set up large referral networks with banks, they maintain quality assurance relationships with many builders, or they get work from energy-efficient mortgage facilitators who have themselves created the necessary referral networks.

Instead of relying solely on diagnostics to bring in revenues, many building performance specialists combine contracting with diagnostics. This strategy increases the gross margin per job and reduces the need to acquire so many customers. Combining contracting with diagnostics can also be a strong selling point for consumers. Consumers would prefer to find a one-stop shop, because it saves them time and allows them to establish trust with one contractor instead of several. If you provide only the diagnostic half of

the solution and leave customers to their own devices for the other half, will they see your service as useful?

Cost Effectively Getting Customers

Comfort Diagnostics in Little Rock, Arkansas, started out just doing diagnostics on buildings, but quickly found out that customers like to pay for work, not for opinions. In a move to combine doing inspections with contracting, they started offering insulation services. When they realized that they were subcontracting out too much HVAC work, they acquired an HVAC contractors license. This enabled them to do large jobs with larger margins. As a differentiated high-quality HVAC contractor, they also had an identity that consumers could recognize and easily locate in the yellow page listings.

Comfort Diagnostics' next challenge was to acquire more customers. Blessed with glib tongues, they turned to radio, and invested heavily in an hour-long talk show that focused on using building science to solve comfort problems and reduce high utility bills. Now the talk show, combined with the resulting word-of-mouth referrals, brings in the bulk of their business. The investment in radio was not cheap; it required a long-term commitment of time and money. But it has paid off in an ongoing stream of customers. Radio is an excellent educational medium, and what Comfort Diagnostics is doing is educating customers through a weekly repetition of practical building science advice and testimonials from customers. Thanks to their rapid adaptability, Comfort Diagnostics has the fastest growing

Opportunities by Climate Zone

Cold Climate: Rotting roofs, drafty rooms, poor radiant comfort, ice dams, mold, wet basements, window condensation, backdrafting water heaters, peeling paint, and high heating-fuel bills.

Hot-Humid Climate: Mold, uncontrolled humidity, unbalanced cooling, poor radiant comfort, backdrafting water heaters, moisture plane issues.

Hot-Dry/Mixed-Dry Climate: Backdrafting water heaters and cooling energy bills.

Mixed Climate: Everything in moderation.

For more information on climatic effects, visit Joe Lstiburek's Web site at www.buildingscience.com.



building performance business that I know about.

Fred Lugano of Lake Construction is another successful building performance contractor who combines diagnostics with roofing and remodeling. Located in Charlotte, Vermont, Fred realized that rotting roofs and ice dams were a big potential market for his services.

But instead of hanging out a shingle as a building performance contractor, Fred invested in yellow page ads in the "Roofing Contractor" section. These ads focused on Lake Construction's ability to solve a variety of roofing-related problems. As Fred puts it, "Sell the benefit, not the service."

Another means of acquiring customers is through contractor-to-contractor referral systems. In Toronto, when a roofer sees a moisture related problem, he or she calls in a building performance specialist and includes the building performance work as a subcontract in the roofing proposal. To provide customer service benefits, the utilities in that area are now sponsoring the development of a trade association to help support this referral process. There are many of these referral networks, generally on a smaller scale, in New York and elsewhere. IAQ referrals are also gaining in numbers. Home performance contractors are linking with healthy indoor air groups and are getting a steady stream of referrals.

Still another profitable type of partnership is a linkup between an HVAC contractor and an insulation contractor. Ed Bishop of Advanced Heating



Pete Collison, of Affordable Energy, and Ed Bishop of Advanced Heating Concepts, look over the boiler system in a house that got the full treatment—from the heating system to the shell.

Concepts and Pete Collison of Affordable Energy, who met at a utility training for whole-house contractors, and who operate outside of Albany, New York, teamed up to share leads and subcontracts. Ed does the HVAC design and installation, while Pete does the diagnostics and the envelope work. The combination lets them offer the performance and cost benefits of a whole house approach. When one sees a job that needs the other's skills, he gets a quote for a subcontract.

Business Development Resources

Once you have a handle on the motivating factors in your local marketplace, and a sense of how you might acquire customers, examine your skills and seek out training in the areas where you are weakest. To learn more about running an independent contracting business, consider enrolling in either of the National Association of Remodelers (NARI) or the National Association of Home Builders (NAHB) Remodelers Council certification programs that cover mostly business issues. For less time-consuming tips on improving an existing operation, I have found the *Small Business Advancement National Cen-*



Frazer Dougherty, of North Fork Retrofit in New York, tests the performance of a retrofit ventilation system with a delivered flow hood (a fixed-plate orifice) that is able to measure small flows.

ter Newsletter to be a helpful resource. To receive this newsletter, send an e-mail to webmaster@www.sbaer.uca.edu with the word "add" in the subject line.

The small-business side of being a HVAC contractor has been explored and well documented by the Air Conditioning Contractors of America (ACCA). They have many resources supporting the development of a professional, quality-driven HVAC business. In addition to business resources, ACCA publishes a number of technical manuals for residential and commercial HVAC design. For example, ACCA's *Manual J* is the reference for residential system sizing, and *Manual D* is the reference for residential duct design. I find the current version of *Manual D* useful when talking to contractors because it is an HVAC industry source that references duct leakage studies and blower door testing. All of these organizations have excellent Web sites.

Another excellent reference for local training opportunities is *Home Energy's* training guide ("Training for Tomorrow: Are Your Contractors Certifiable?" *HE*, Jan/Feb '99, p. 29). This guide is also available on the Web. (*Home Energy's* Web site is generally a great place to get information, and I often tell potential customers to visit it.) I also use the *Home Performance Contractor* brochure and the *No Regrets Remodeling* book, both published by *Home Energy*, as third-party educational materials for consumers.

Sales: Open with Comfort and Close with Efficiency

Whether you choose the company route or the independent-consultant route, you must be comfortable with the sales process. The sales process is about building trust. Use your building science knowledge to initiate the process, and use third-party sources of information—such as Energy Star program materials, the *Home Performance Contractor* brochure, or reprints from *Home Energy*—to support your assertions and consolidate your relationship with your customer.

Despite all the public funding spent to promote energy efficiency, most homeowners still equate saving energy with living in a cold, dark house. Rather than trying to push energy savings "predictions" on an unwilling customer, try emphasizing the non-energy benefits of building science. Customers have seen predictions come and go, and it will be very hard to make them believe in yours. But by documenting the symptoms of building failure and providing clear explanations and third party support for your explanations, it is very possible to convert a customer into a believer in building science and performance testing. Then you hit them with the close: "And it saves energy and pays for itself!"

Shopping for Solutions

I have used this strategy numerous times to convince potential customers to sign a contract with me. In one such case, a condominium association that had experienced \$80,000 in ice damage over a five year period was desperately looking for a solution. They had hired architects and engineers. They had put in added insulation and attic venting. Nothing had worked. Finally, their insurance was canceled because of the recurring ice damage. Through a utility referral process, they contacted a couple of local building performance contractors that I team up with from time to time.

Working with one of the contractors, I went into the building and took pictures of all the usual suspects: attic duct leaks, drop ceilings, pocket doors, lighting fixtures, plumbing chases, and so on. Also, we found that the gas vent for the downstairs apartments was actually run through the boxed-in cold-air return of the upstairs apartments. We went back and worked up our estimate. There was a time when I would have written a long report that included a sophisticated savings estimate, but I stopped doing that when I discovered that people want to pay for the work, not for a long report. The summary report I gave to the condominium association was mostly a bulleted list of the problem areas and an

attachment of photos mounted on card stock with nice captions.

We made our presentation, and passed around the photos. We answered a few questions about the ice on the buildings, and then they asked, "Will we save money?" I said, "Yes," and the job was ours. No time spent on savings calculations. (We did look at the billing data to score the energy intensity of the buildings in Btu/ft² per heating degree day.) Still, they were not ready to buy until there was an economic rationalization, however weak, for doing so. They didn't call to save energy dollars, but saving energy dollars made the investment smart and politically safe. The same logic applies to people making decisions at home. We make them feel smart when we combine the motivating creature comforts of satisfactory indoor temperatures, good indoor air quality, and durable buildings with the economic rationalization of energy savings.

Untapped Market Exists

I firmly believe that there will eventually be a long term consumer driven market for applied building science and performance testing. Studies by Consumers Gas in Toronto and by *Contracting Business* magazine in the early '90s found that 10% or so of homes have building performance-related problems that their owners want to solve now. There should be plenty of business in that 10% of homes to keep many home performance contractors very busy.

Furthermore, as Energy Star and green building programs get more public recognition, more and more builders will require the services of building performance subcontractors. I witnessed this phenomenon firsthand at the first National Green Building Conference put on by the NAHB Research Center. There I heard builder after builder get up and say that he or she had gone green and was pleased to

Fitting into the Marketplace

Contractors can be divided into three groups based on how they use building science and performance testing in their business. These categories make it easier to target building performance marketing and training programs to reach specific kinds of contractors.

The Building Performance Contractor

The highly differentiated, problem solver contractor is the one customers yearn for when they have a nasty building performance problem that nobody else can solve, or when they really want to ensure a high-performance building. Unfortunately, customers have to seek high and low to find such a contractor, and in many areas no such contractor exists. Start the search by contacting organizations that fund contractor trainings and related programs, such as utility companies, weatherization agencies, energy rating organizations, and state energy offices. This type of contractor lives by his or her reputation and depends on a strong referral network.

The Top-Quality Trade Contractor

The second type of contractor integrates building performance into his or her existing trade, be it HVAC, insulation, roofing, remodeling, or new construction. This contractor uses building science and performance testing to improve the qual-

ity and scope of the work that he or she already does. The real distinction between this type of contractor and the building performance contractor is in how they get their customers. Customers can find the second type of contractor using listings in the yellow pages for the already-established trades. Once the customer has made contact, these contractors use building performance techniques to differentiate themselves from other contractors. Because they offer a recognized service, they can transform customers who call them for roofing, for example, into customers for building performance services.

Top-Quality Builders and Remodelers

Builders and remodelers typically do not want to train their own crews to undertake specialized tasks, especially ones that require a high level of quality control. It is easier to hire a subcontractor. For builders and remodelers who are educated on the benefits of building science and performance testing, a local subcontractor with building performance skills is worth his or her weight in gold (OK, maybe dense-pack cellulose). As Energy Star and green building programs get more public recognition, these builders will require more and more building performance subcontractors.



be both doing the right thing and making more money by selling more houses more quickly. These builders' big problem now was finding qualified insulation, HVAC, and blower door testing subcontractors who could do the building performance work required to make their green buildings function

well. Some of these builders had been driven to the extreme of capitalizing and training new subcontractors because their existing subs were unwilling to make the switch. Now that's dedication, and it offers a real opportunity for subcontractors with building performance skills.

Training and Certification

In 1979, I put my engineering background to work helping a friend who was starting a business installing solar hot water heaters and wood boilers. While researching how to save energy, I came across information on blower doors and started reading the ASHRAE transactions on measuring air leakage in buildings. I built a blower door and began offering "Whole House Audits" as a way to differentiate our heating company from other contractors. We created a relationship with an insulation contractor to offer a complete HVAC and air sealing package.

After a few years of selling and installing furnaces and doing air sealing, I ran into Larry Kinney, of Syntech Systems, who was building a

blower door and doing DOE low income weatherization training and technical assistance work. I ended up working for Larry and was the lead trainer for introducing blower doors into the Pennsylvania and New York weatherization programs.

In 1986, I went off to restart a dormant not-for-profit energy service company, Syracuse Energy or SyrESCO. We did a lot of work with not-for-profits and multifamily buildings. During this time, I also got my heating contractors license. In 1993 when I left, there were over 20 employees. Since leaving the not-for-profit, I have completed a myriad of energy efficiency consulting projects, including program, software, curriculum, and association development.

There is just too much value in the services we offer for the market to ignore them much longer. The ability to take control of the movement of air, the transport of moisture, and the transfer of heat in buildings allows us to provide high-quality, healthy living environments—and we save energy for our clients. With an ever-increasing concern for health and comfort, how can we lose? Those people who combine building performance expertise with good business skills will have a head start in mining or creating that market in their own backyard. Good luck!

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